SIH26143 is partially solved: oil-spill detection, drift, and AIS correlation exist, but **fully operational, explainable vessel attribution** remains incomplete.

Existing Systems

The closest existing systems already cover large parts of the pipeline, especially **satellite detection**, **drift forecasting**, and **GIS delivery**. Europe's CleanSeaNet is the strongest operational benchmark: it provides satellite-based oil-spill and vessel detection for EU waters, uses SAR scenes, integrates AIS from SafeSeaNet, and has added drift analysis for source estimation (Hajduch, 2021). CleanSeaNet is operational rather than experimental, and alerts are used to support legal and operational follow-up in national EEZs (Carpenter, 2015; Santos et al., 2024).

Several research systems now replicate large parts of this architecture with Sentinel-1 and AI. A near-real-time Mediterranean system built around Sentinel-1 includes five subsystems: satellite processing, oil-spill detection, synoptic forecasting, oil slick trajectory simulation, and web interface (Yang et al., 2024). A related Copernicus-based system uses a two-step deep-learning detector plus segmentation and reports about six minutes from two Sentinel-1 scenes to binary masks (Yang & Schnupfhagn, 2025). Open-source and low-cost operational systems have also been demonstrated in Egypt and the Gulf of Mexico using Sentinel-1 plus GIS-based alerting and contextual layers (El-Magd et al., 2021; Uribe-Martínez et al., 2024).

Title: Closest Existing Systems

System	Satellite	AI	AIS	Drift	Attribution	GIS	Similarity to SIH26143
EMSA CleanSeaNet	Yes	Limited public detail	Yes	Yes	Partial	Yes	**Very high** (Hajduch, 2021)
ISTOP + probabilistic vessel association	Yes	Not central	Yes	Yes	Yes	Yes	**Very high** (Busler et al., 2015)
Mediterranean NRT warning system	Yes	Yes	No public AIS layer stated	Yes	No	Yes	High (Yang et al., 2024)
Luo 2024 ship tracing system	Yes	Limited	Yes	Yes	Yes	Not emphasized	**Very high** (Luo et al., 2024)

System	Satellite	AI	AIS	Drift	Attribution	GIS	Similarity to SIH26143
Chang 2024 SAR + AIS tracking	Yes	Yes	Yes	Implicit only	Partial	No	High (Chang et al., 2024)

FIGURE 1 Closest existing systems to the SIH26143 pipeline

Open-source building blocks also exist, but usually as components rather than complete products. OpenDrift is an operational open-source Lagrangian framework with built-in backtracking and oil-drift capability (Dagestad et al., 2017). RIOSS is an open-source SAR oil-spill detection stack using random forests, with planned augmentation from Sentinel-3 wind and Copernicus Marine currents (Conceição et al., 2021). MIDLN has public code for interpretable SAR oil-spill identification (Fan et al., 2024).

Indian Ecosystem

India clearly has **drift advisory and maritime awareness capability**, but the evidence for a public **end-to-end oil-spill attribution platform** is limited. INCOIS already provides operational ocean information services supported by numerical modelling, data assimilation, and infrastructure for dissemination (Kumar & Francis, 2024). More specifically, Indian Coast Guard responders already use INCOIS oil-spill advisory services, and INCOIS has applied GNOME with Indian ocean-current products to Ennore, MV Rak, and Mauritius events (Prasad et al., 2022; Prasad et al., 2018; Prasad et al., 2020; Prasad et al., 2022).

Indian work on satellite detection is substantial. Sentinel-1 SAR has been used for Ennore/Chennai oil spill mapping, with VV found more useful than VH in that case, and supervised classifiers reaching high reported accuracy (Dasari et al., 2021). A separate Indian Ocean study used Sentinel-1 with GNOME and ADIOS and concluded SAR is reliable under moderate wind conditions, though near-real-time processing is computationally demanding (Naz et al., 2020).

India's broader maritime surveillance backbone is real but focused on MDA, not pollution attribution. The national MDA architecture includes coastal AIS receivers, radars, and the Information Management and Analysis Centre feeding a common operating picture to Navy and Coast Guard nodes (Baruah, 2016). IFC-IOR further extends regional information fusion for safety and security (Kumar, 2024). That means India has **pieces of the pipeline in different institutions**, but the supplied evidence does not show a public NTRO/ISRO/NRSC/DRDO system that already fuses satellite spill detection, drift hindcasting, AIS correlation, vessel ranking, and explainable attribution into one operational platform.

Title: Indian Capability Map

Layer	Indian Evidence	Status
SAR spill detection	Sentinel-1 applied at Chennai and Indian Ocean (Dasari et al., 2021; Naz et al., 2020)	Present
Drift advisory	INCOIS + GNOME + HOOFS / ensembles (Prasad et al., 2022; Prasad et al., 2020)	Operational

Layer	Indian Evidence	Status
AIS / MDA fusion	IMAC, coastal AIS chain, IFC-IOR (Baruah, 2016; Kumar, 2024)	Operational
Vessel attribution from spill	No public Indian end-to-end system in corpus	Unclear / likely partial
Explainable forensic dashboard	No direct Indian evidence in corpus	Missing

FIGURE 2 Indian public evidence across the SIH26143 stack

International Ecosystem

Internationally, the landscape is mature in **detection** and increasingly strong in **combined spill-plus-ship analysis**. Sentinel-1 changed the field by making free NRT SAR available and enabling semiautomatic operational monitoring at scale (Topouzelis & Singha, 2016). Large operational and research datasets have since enabled deep learning systems using DeepLab, U-Net variants, Faster R-CNN, Mask R-CNN, and hybrid CNN-transformer models (Krestenitis et al., 2019; Huang et al., 2022; Yekeen et al., 2020; Dehghani-Dehcheshmeh et al., 2023).

There is also clear progress toward attribution. Busler's ISTOP-linked system combined AIS, LRIT, SAR ship detections, spill detections, and drift models in a probabilistic association framework (Busler et al., 2015). Luo 2024 went further by filtering AIS near spills, converting slick areas into trajectory points, and ranking likely culprit ships using bidirectional drift similarity (Luo et al., 2024). Chang 2024 combined improved U-Net spill and ship detection with AIS trajectory screening based on proximity and parallelism to the spill extension (Chang et al., 2024). Zhao 2025 also reports successful identification of spill sources by combining multiclass YOLOv8 detection with AIS (Zhao et al., 2025).

Drift and source reconstruction are advanced but still not settled. MEDSLIK-II, OpenDrift, and GNOME are widely validated, often against SAR or drifters, and can run forecast or hindcast modes (Dominicis et al., 2013; Dagestad et al., 2017; Pärt et al., 2023; Prasad et al., 2020). Bayesian source reconstruction from spill contours is now a research topic, including estimation of source location, time, duration, and released quantity, often with uncertainty outputs (Mohtar et al., 2021; Mohtar et al., 2024). But naive backtracking can fail when drift properties change or current fields are biased, so attribution remains probabilistic rather than definitive (Breivik et al., 2025; De Meutter et al., 2021).

Title: International Maturity by Capability

Capability	Maturity	Evidence
Sentinel-1 oil detection	High	Operational and research systems widespread (Topouzelis & Singha, 2016; Yang et al., 2022)
AI spill segmentation	High	Many strong DL models exist (Krestenitis et al., 2019; Xiaoshuang et al., 2022)

Capability	Maturity	Evidence
Drift forecasting / hindcast	High	Mature operational models (Zodiatis et al., 2012; Dagestad et al., 2017)
AIS-based suspect screening	Moderate-high	Demonstrated in several studies (Busler et al., 2015; Chang et al., 2024)
Explainable attribution	Low-moderate	Emerging, not standard (Murugan et al., 2026; Fan et al., 2024)

FIGURE 3 International maturity across major capability layers

Commercial and Open-Source Overlap

The strongest commercial-style overlap in the supplied corpus comes from **operational maritime surveillance platforms**, not pure oil-spill systems. SatShipAI is an operational AI platform using Sentinel-1 SAR and AIS for long-term maritime surveillance around Denmark, including dark-vessel use cases (Nasios & Vogklis, 2025). OSIRIS-like maritime traffic platforms integrate satellite imagery, AIS, ship detection, route prediction, and WebGIS, but target broader illegal traffic monitoring rather than oil-spill attribution specifically (Reggiannini et al., 2024). Antasena similarly integrates AIS, satellite imagery, anomaly detection, and a smart dashboard for maritime security, with SAR integration still under development (Badrudin et al., 2026).

The open-source ecosystem is better at components than complete end-to-end pipelines. OpenDrift covers transport modeling (Dagestad et al., 2017). MIDLN provides interpretable SAR detection code (Fan et al., 2024). RIOSS provides open SAR classification routines (Conceição et al., 2021). Earlier GIS cloud tools and decision-support systems show that web delivery is not novel on its own (Fustes et al., 2012; Fetissov et al., 2021).

Top 5 Competitors / Existing Systems

- **EMSA CleanSeaNet**: operational satellite oil-spill and vessel detection with AIS and drift-supported source estimation (Hajduch, 2021).
- **ISTOP probabilistic vessel association system**: Canadian system linking spill detections, AIS/LRIT/SAR vessel tracks, and drift in time-space (Busler et al., 2015).
- **Luo 2024 multi-source ship tracing**: the closest research analogue to SIH26143's attribution core (Luo et al., 2024).
- **Mediterranean Sentinel-1 NRT warning systems**: operational AI detection plus trajectory simulation plus web interface (Yang et al., 2024).
- **SatShipAI / OSIRIS-class maritime surveillance platforms**: broader SAR+AIS+AI+WebGIS surveillance overlap (Nasios & Vogklis, 2025; Reggiannini et al., 2024).

Similarity Analysis and Gaps

Potential overlap exists. The combination of **SAR spill detection + AIS filtering + drift modelling + source ranking** has already been built in research form and, in parts, in operational government services (Luo et al., 2024; Busler et al., 2015; Hajduch, 2021). That means SIH26143 is **not** an untouched idea.

What is not clearly solved is the **same complete pipeline as a transparent, operational, public, explainable vessel-attribution platform**. Existing systems usually stop at one of four points: detection only, drift only, AIS-only anomaly screening, or non-explainable suspect identification (Blondeau-Patissier et al., 2023; Yekeen & Balogun, 2020; Abbas et al., 2025; Zhao et al., 2025). Explainability is still emerging in SAR oil-spill AI, and uncertainty-aware source reconstruction remains more research-grade than routine operational practice (Murugan et al., 2026; Mohtar et al., 2021; Breivik et al., 2025).

Title: Where Overlap Ends

Pipeline Stage	Already Exists?	Main Missing Piece
SAR spill detection	Yes	Generalization across regions
AIS correlation	Yes	Robust handling of missing/dark AIS
Drift hindcasting	Yes	Stable uncertainty-aware origin estimation
Vessel attribution	Partial	Explainable ranked evidence
Full public end-to-end platform	Partial	Integrated operational transparency

FIGURE 4 Where current overlap ends in the full pipeline

Top 5 Existing Limitations

- **Look-alikes still cause false alarms** in SAR and no single sensing mode solves this cleanly (Krestenitis et al., 2019; Wan et al., 2026; Yekeen & Balogun, 2020).
- **Drift modeling remains uncertainty-sensitive** to currents, winds, and parameterization (Dominicis et al., 2013; Accarino et al., 2025; De Meutter et al., 2021).
- **AIS is incomplete or degraded**, weakening attribution confidence (Liang et al., 2024; Jayaswal & Ghosh, 2025).
- **Operational systems are often closed or partially opaque**, especially government services (Hajduch, 2021; Santos et al., 2024).
- **Most platforms optimize surveillance or response, not explainable forensic attribution** (Fetissov et al., 2021; Reggiannini et al., 2019; Badrudin et al., 2026).

Top 5 Possible Differentiators for Your Team

- **Explainable suspect ranking**, because explainability is still rare in oil-spill pipelines (Murugan et al., 2026).
- **Explicit uncertainty display** for origin and vessel responsibility, which current drift literature says is crucial but under-implemented (Breivik et al., 2025; De Meutter et al., 2021).
- **Indian operational context integration**, especially INCOIS current products and Coast Guard workflows (Prasad et al., 2022; Baruah, 2016).
- **Better dark/missing AIS handling**, using broader maritime intelligence ideas rather than assuming cooperative vessels (Jayaswal & Ghosh, 2025; Nasios & Vogklis, 2025; Galdelli et al., 2025).
- **A single case-centric forensic dashboard** spanning spill mask, drift, candidate tracks, and rationale, rather than disconnected tools (Reggiannini et al., 2024; Fetissov et al., 2021; Fustes et al., 2012).

Final Conclusion

SIH26143 is **partially solved**, not genuinely blank. The individual modules—Sentinel-1 spill detection, AI classification, GNOME/OpenDrift/MEDSLIK drift simulation, AIS anomaly analysis, and GIS dashboards—are all well established, and several systems already combine multiple modules into near-end-to-end workflows (Topouzelis & Singha, 2016; Krestenitis et al., 2019; Dagestad et al., 2017; Busler et al., 2015; Fetissov et al., 2021). There is real **potential overlap** with CleanSeaNet-class services, ISTOP-style probabilistic association, and newer multi-source ship-tracing research (Hajduch, 2021; Busler et al., 2015; Luo et al., 2024).

The biggest shift in the evidence is that the field has moved from **oil-spill detection alone** to **integrated detection-plus-attribution pipelines**. The biggest open question is whether any team can make that pipeline **operationally credible, uncertainty-aware, and explainable enough** that authorities would trust it for real suspect-vessel investigation rather than only as a screening aid.

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